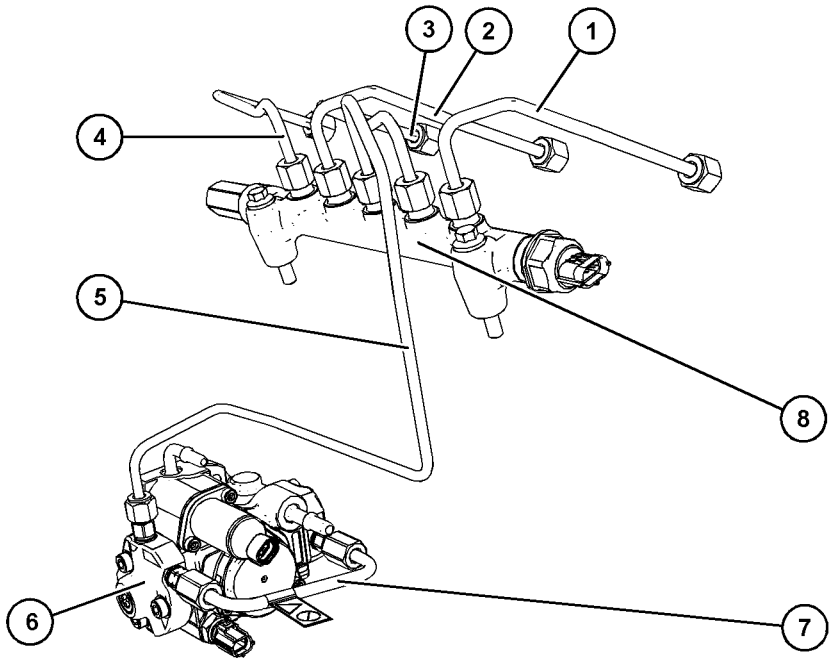


Illustration 24 g06334696

(1) High-pressure line
(2) High-pressure line
(3) High-pressure line
(4) High-pressure line
(5) Fuel transfer line that is high pressure
(6) High-pressure pump
(7) High-pressure line
(8) High-pressure fuel manifold (rail)

The high-pressure fuel lines are the fuel lines that are between the high-pressure fuel pump and the high-pressure fuel manifold and the fuel lines that are between the fuel manifold and electronic unit injectors. These fuel lines are different from fuel lines on other fuel systems.

The fuel lines are different because of the following items:

- The high-pressure fuel lines are constantly charged with high pressure.
- The internal pressures of the high-pressure fuel lines are higher than other types of fuel system.
- The high-pressure fuel lines are formed to shape and then strengthened by a special process.

Do not step on the high-pressure fuel lines. Do not deflect the high-pressure fuel lines. Do not bend or strike the high-pressure fuel lines. Deformation or damage of the high-pressure fuel lines may cause a point of weakness and potential failure.

Do not check the high-pressure fuel lines with the engine or the starting motor in operation. After the engine has stopped, you must wait for 10 minutes to allow the fuel pressure to be purged from the high-pressure fuel lines before any service or repair is performed on the engine fuel lines. The 10 minute wait will also allow static charge to dissipate from the low-pressure fuel system.

Do not loosen the high-pressure fuel lines to remove air from the fuel system. This procedure is not required.

Visually inspect the high-pressure fuel lines before the engine is started. This inspection should be each day.

If you inspect the engine in operation, always use the proper inspection procedure to avoid a fluid penetration hazard. Refer to Operation and Maintenance Manual, “General hazard Information”.

- Inspect the high-pressure fuel lines for damage, deformation, a nick, a cut, a crease, or a dent.

- Do not operate the engine with a fuel leak. If there is a leak, do not tighten the connection, to stop the leak. The connection must only be tightened to the recommended torque. Refer to Disassembly and Assembly, “Fuel injection lines - Remove and Fuel injection lines - Install”.
- If the high-pressure fuel lines are torqued correctly and the high-pressure fuel lines are leaking, the high-pressure fuel lines must be replaced.
- Ensure that all clips on the high-pressure fuel lines are in place. Do not operate the engine with clips that are damaged, missing, or loose.
- Do not attach any other item to the high-pressure fuel lines.
- Loosened high-pressure fuel lines must be replaced. Also removed high-pressure fuel lines must be replaced. Refer to Disassembly and assembly manual, “Fuel Injection Lines - Install”.

i07728672

Before Starting Engine

SMCS Code: 1000

NOTICE
For initial start-up of a new or rebuilt engine, and for start-up of an engine that has been serviced, make provision to shut the engine off should an overspeed occur. This may be accomplished by shutting off the air and/or fuel supply to the engine.

⚠ WARNING

Engine exhaust contains products of combustion which may be harmful to your health. Always start and operate the engine in a well ventilated area and, if in an enclosed area, vent the exhaust to the outside.

Inspect the engine for potential hazards.

Do not start the engine or move any of the controls if there is a “DO NOT OPERATE” warning tag or similar warning tag attached to the start switch or to the controls.

Before starting the engine, ensure that no one is on, underneath, or close to the engine. Ensure that the area is free of personnel.

If equipped, ensure that the lighting system for the engine is suitable for the conditions. Ensure that all lights work properly, if equipped.

All protective guards and all protective covers must be installed if the engine must be started to perform service procedures. To help prevent an accident that is caused by parts in rotation, work around the parts carefully.

Do not bypass the automatic shutoff circuits. Do not disable the automatic shutoff circuits. The circuits are provided to help prevent personal injury. The circuits are also provided to help prevent engine damage.

Before starting the engine, ensure that the drive belt is securely in place. The coolant pump is operated by the drive belt.

See the Service Manual for repairs and for adjustments.

i08219601

Engine Starting

SMCS Code: 1000

⚠ WARNING

Do not use aerosol types of starting aids such as ether. Such use could result in an explosion and personal injury.

If a warning tag is attached to the engine start switch or to the controls DO NOT start the engine or move the controls. Consult with the person that attached the warning tag before the engine is started.

All protective guards and all protective covers must be installed if the engine must be started to perform service procedures. To help prevent an accident that is caused by parts in rotation, work around the parts carefully.

Start the engine from the operator's compartment or from the engine start switch.

Always start the engine according to the procedure that is described in the Operation and Maintenance Manual, “Engine Starting” topic in the Operation Section. Knowing the correct procedure will help to prevent major damage to the engine components. Knowing the procedure will also help to prevent personal injury.

To ensure that the jacket water heater (if equipped) and/or the lube oil heater (if equipped) is working correctly, check the water temperature gauge. Also, check the oil temperature gauge during the heater operation.

⚠ WARNING

Engine exhaust contains products of combustion which may be harmful to your health. Always start and operate the engine in a well ventilated area and, if in an enclosed area, vent the exhaust to the outside.

Note: The engine is equipped with a device for cold starting. If the engine will be operated in very cold conditions, an extra cold starting aid may be required. Normally, the engine will be equipped with the correct type of starting aid for the region of operation.

These engines are equipped with a glow plug starting aid in each individual cylinder that heats the intake air to improve starting. Some Caterpillar engines may have a cold starting system that is controlled by the ECM that allows a controlled flow of ether into the engine. The ECM will disconnect the glow plugs before the ether is introduced. This system would be installed at the factory.

i02234873

Engine Stopping

SMCS Code: 1000

Stop the engine according to the procedure in the Operation and Maintenance Manual, “Engine Stopping (Operation Section)” in order to avoid overheating of the engine and accelerated wear of the engine components.

Use the Emergency Stop Button (if equipped) ONLY in an emergency situation. Do not use the Emergency Stop Button for normal engine stopping. After an emergency stop, DO NOT start the engine until the problem that caused the emergency stop has been corrected.

Stop the engine if an overspeed condition occurs during the initial start-up of a new engine or an engine that has been overhauled.

To stop an electronically controlled engine, cut the power to the engine and/or shutting off the air supply to the engine.

i08154895

Electrical System

SMCS Code: 1000; 1400

Never disconnect any charging unit circuit or battery circuit cable from the battery when the charging unit is operating. A spark can cause the combustible gases that are produced by some batteries to ignite.

To help prevent sparks from igniting combustible gases that are produced by some batteries, the negative “−” cable should be connected last from the external power source to the primary position for grounding.

Check the electrical wires daily for wires that are loose or frayed. Tighten all loose electrical connections before the engine is started. Repair all frayed electrical wires before the engine is started. See the Operation and Maintenance Manual for specific starting instructions.

Grounding Practices

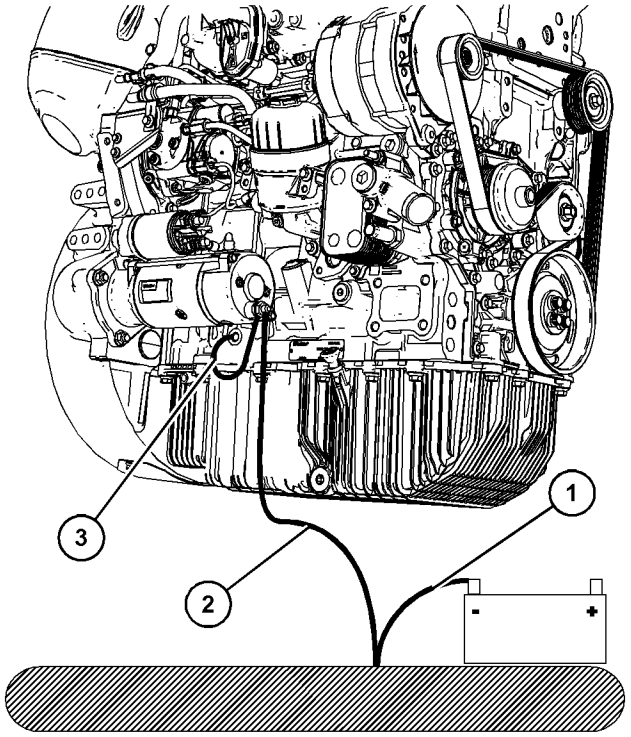


Illustration 25 g06296580

Typical example

(1) Ground to battery
(2) Ground to starting motor
(3) Starting motor to engine block

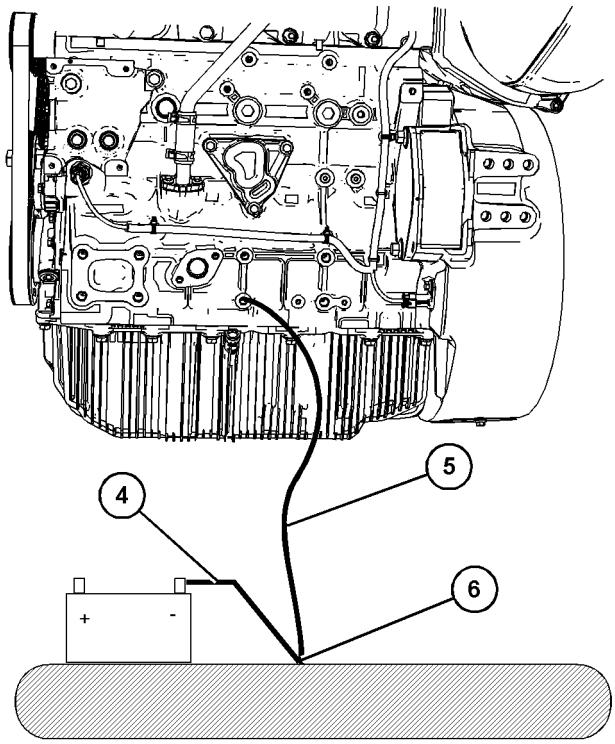


Illustration 26 g06296637

Typical example

(4) Ground to the battery
(5) Ground to the engine block
(6) Primary position for grounding

NOTICE

For C3.6 industrial engines, ensure that a 12 VDC or 24 VDC battery source is used to start the engine. For C2.8 industrial engines, use a 12 VDC battery source to start the engine. Never attempt to start an engine from an external power source such as electric welding equipment, which has a voltage that is unsuitable for engine starting and will damage the electrical system.

Correct grounding for the engine electrical system is necessary for optimum engine performance and reliability. Incorrect grounding will result in uncontrolled electrical circuit paths and in unreliable electrical circuit paths.

Uncontrolled electrical circuit paths can result in damage to the crankshaft bearing journal surfaces and to aluminum components.

Engines that are installed without engine-to-frame ground straps can be damaged by electrical discharge.

To ensure that the engine and the engine electrical systems function correctly, an engine-to-frame ground strap with a direct path to the battery must be used. This path may be provided by way of a direct engine ground to the frame.

The connections for the grounds should be tight and free of corrosion. The engine alternator must be grounded to the negative “-” battery terminal. The wire used must be adequate to handle the full charging current of the alternator.

The power supply connection for the engine electronics should always be from the isolator to the battery . It is the decision of the OEM which polarity the isolator is connected to.

i07925886

Engine Electronics

SMCS Code: 1000; 1900

⚠ WARNING

Tampering with the electronic system installation or the OEM wiring installation can be dangerous and could result in personal injury or death and/or engine damage.

⚠ WARNING

Electrical Shock Hazard. The electronic unit injectors use DC voltage. The ECM sends this voltage to the electronic unit injectors. Do not come in contact with the harness connector for the electronic unit injectors while the engine is operating. Failure to follow this instruction could result in personal injury or death.

This engine has a comprehensive, programmable Engine Monitoring System. The Electronic Control Module (ECM) has the ability to monitor the engine operating conditions. If any of the engine parameters extend outside an allowable range, the ECM will initiate an immediate action.

The following actions are available for engine monitoring control:

- Warning: A warning light could be illuminated or a sound warning could be activated (horn)
- Derate: The engine power could be reduced by 50 percent
- Shutdown: The engine could be shut down or only operate at low idle

The following monitored engine operating conditions and components can limit engine speed and/or the engine power:

- Engine Coolant Temperature
- Engine Oil Pressure
- Engine Speed
- Intake Manifold Air Temperature
- Intake Manifold Pressure
- Wastegate Regulator
- Supply Voltage to Sensors
- Fuel Temperature
- Fuel Pressure in Manifold (Rail)
- NOx Reduction System temperature
- Engine Aftertreatment System

The Engine Monitoring package can vary for different engine models and different engine applications. However, the monitoring system and the engine monitoring control will be similar for all engines.

Note: Many of the engine control systems and display modules that are available for Caterpillar engines will work in unison with the Engine Monitoring System. Together, the two controls will provide the engine monitoring function for the specific engine application. Refer to Troubleshooting for more information on the Engine Monitoring System.

Product Information Section

General Information

i07853978

Model View Illustrations

SMCS Code: 1000

The following model views show typical features of the engine. Due to individual applications, your engine may appear different from the illustrations.

C3.6 Industrial Engine Views

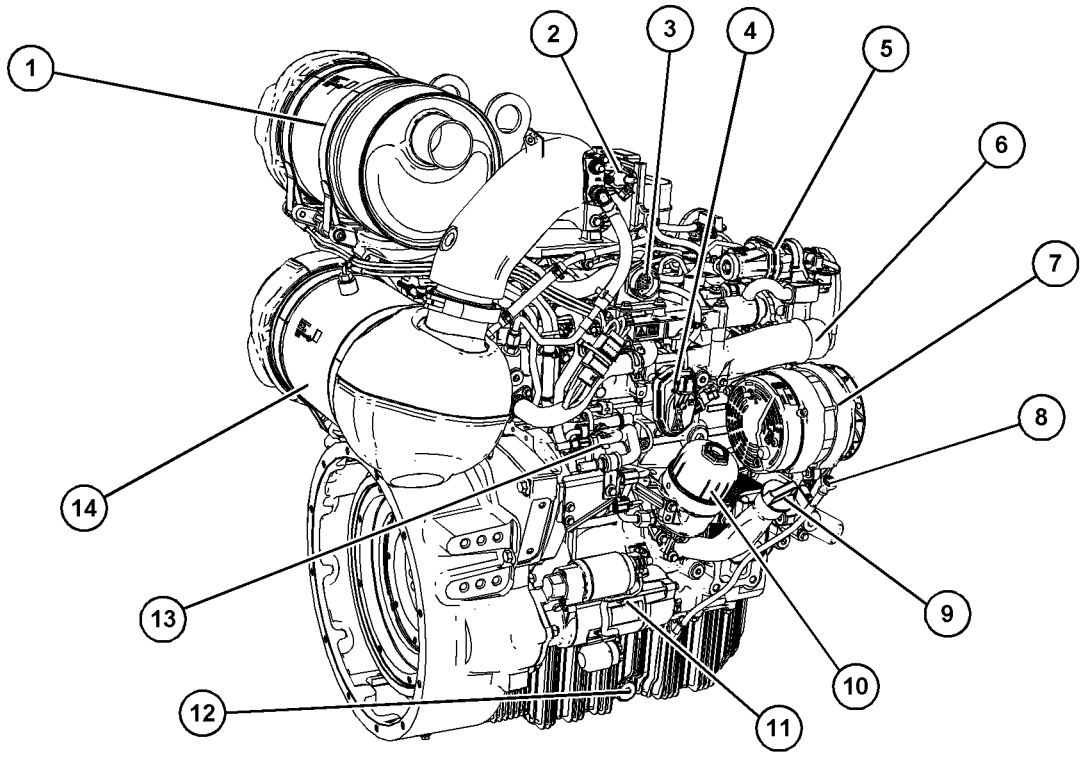


Illustration 27 g06297496

Typical example

(1) Selective Catalytic Reduction (SCR)
(2) Diesel Exhaust Fluid (DEF) injector
(3) Engine harness interface
(4) Throttle control valve
(5) NOx Reduction System (NRS) valve
(6) Air intake from air charge cooler
(7) Alternator
(8) Oil gauge (Dipstick)
(9) Oil filler (lower)
(10) Oil filter assembly
(11) Starting motor
(12) Oil drain plug
(13) High-pressure fuel pump
(14) Diesel Particulate Filter (DPF)

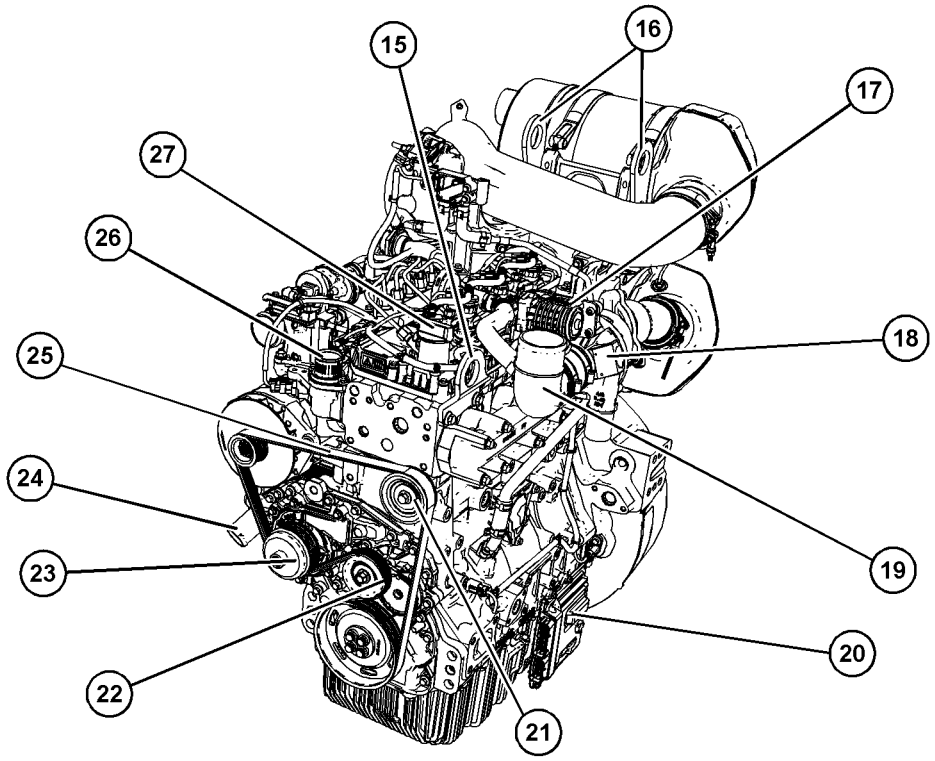


Illustration 28 g06297569

Typical example

(15) Front lifting eye
(16) Rear lifting eyes
(17) Actuator for turbocharger
(18) Turbocharger
(19) Air intake from air cleaner
(20) Electronic Control Module (ECM), location for transportation only
(21) Idler for drive belt
(22) Adjuster for drive belt
(23) Coolant pump pulley
(24) Coolant intake
(25) Drive belt
(26) Coolant outlet
(27) Top oil filler

C2.8 Industrial Engine Views

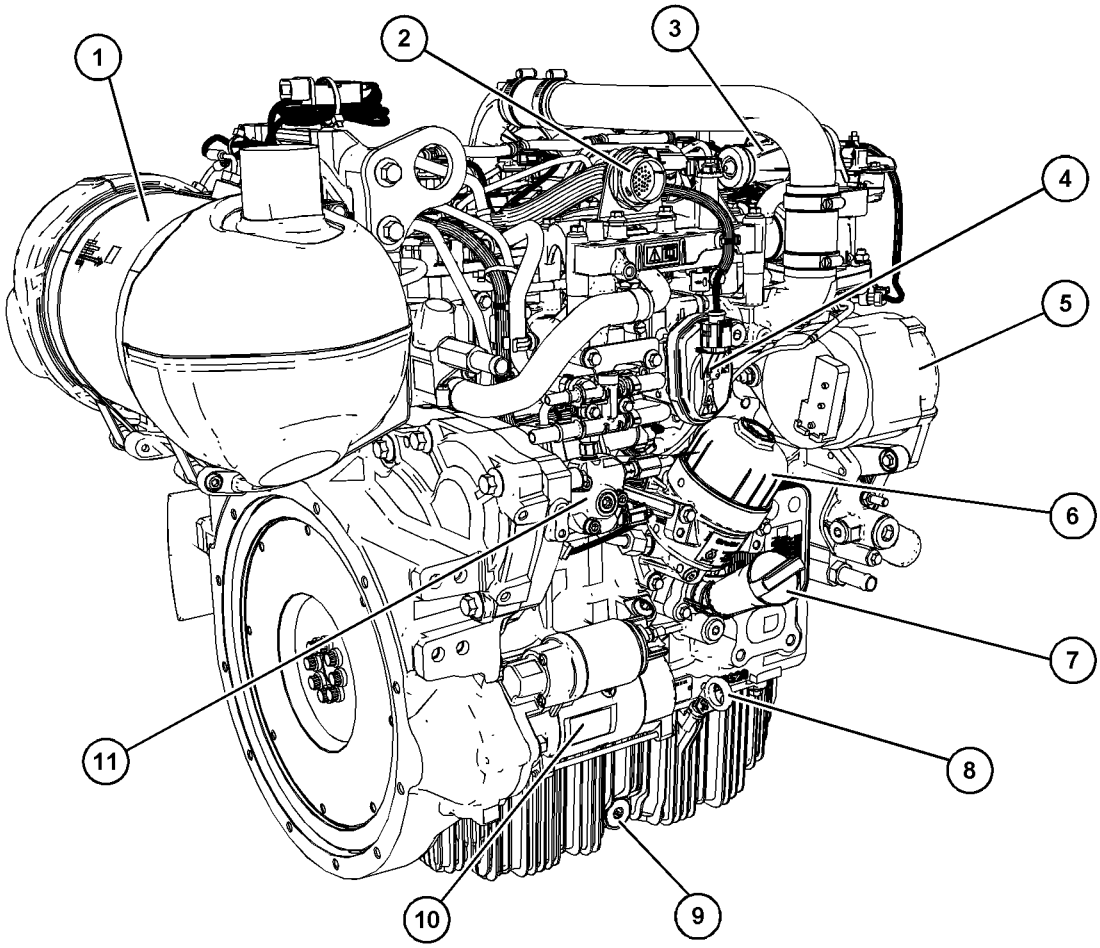


Illustration 29 g06481332

Typical example

(1) Diesel Particulate Filter (DPF) and Diesel Oxidation Catalyst (DOC) combined
(2) Engine harness interface
(3) NOx Reduction System (NRS) valve
(4) Throttle control valve
(5) Alternator
(6) Oil filter assembly
(7) Oil filler (lower)
(8) Oil gauge (Dipstick)
(9) Oil drain plug
(10) Starting motor
(11) High-pressure fuel pump

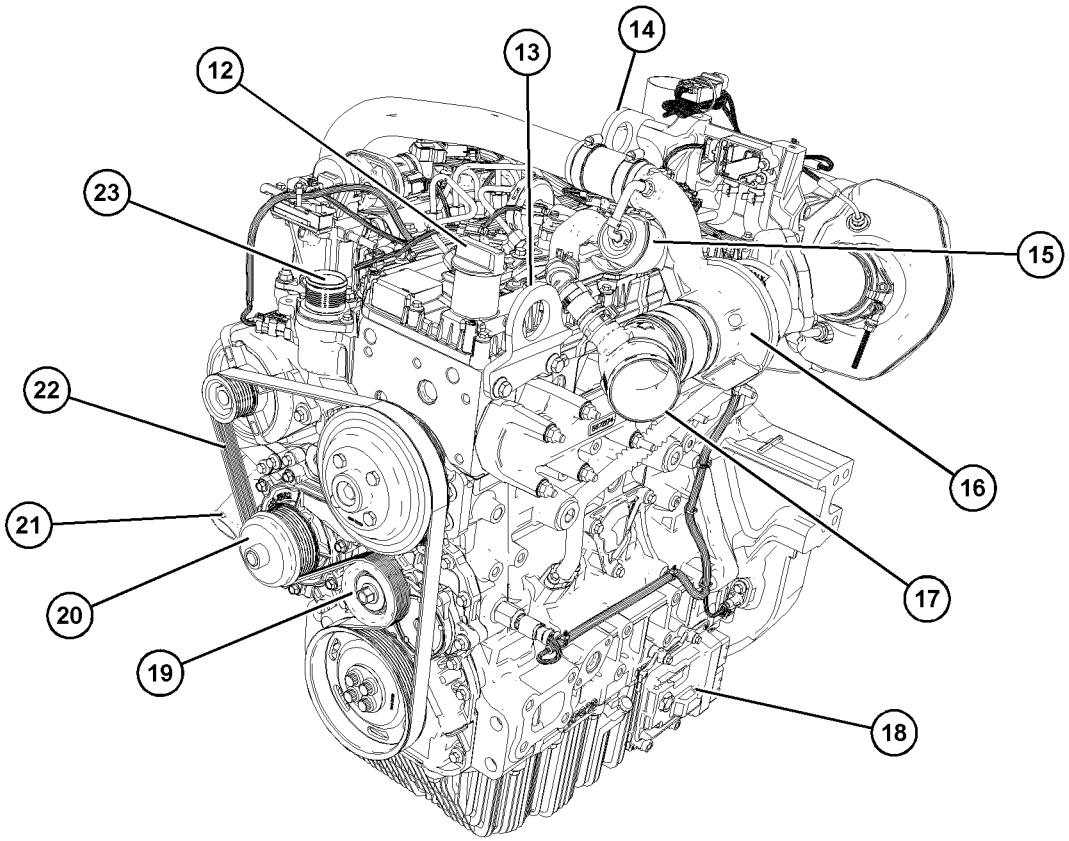


Illustration 30 g06481335

Typical example

(12) Oil filler (upper)
(13) Front lifting eye
(14) Rear lifting eyes
(15) Actuator for turbocharger
(16) Turbocharger
(17) Air intake from air cleaner
(18) Electronic Control Module (ECM), location for transportation only
(19) Adjuster for drive belt
(20) Coolant pump pulley
(21) Coolant intake
(22) Drive belt
(23) Coolant outlet

Loose or Off Engine Components

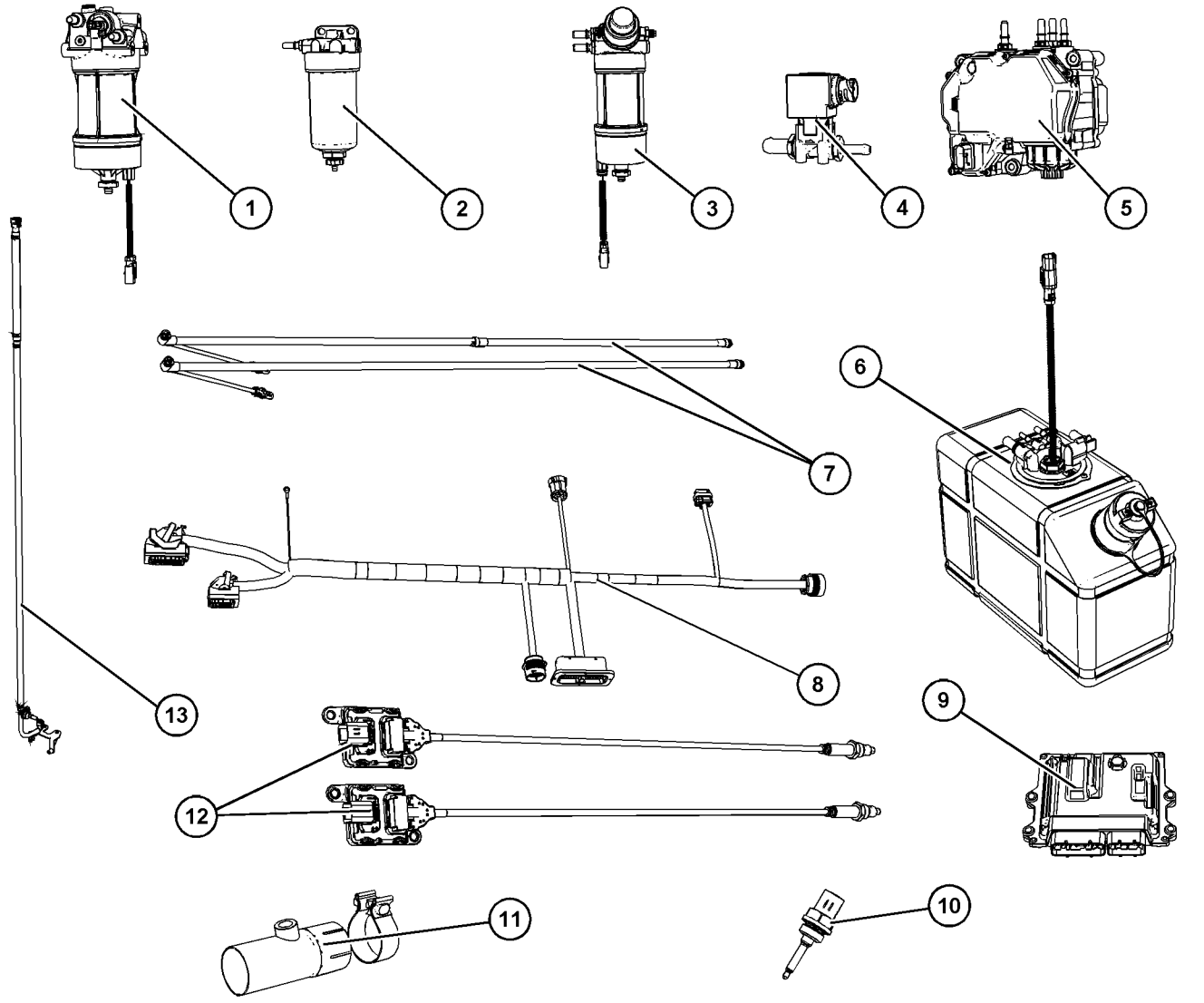


Illustration 31 g06297651

Typical example

(1) Primary fuel filter, combined with electric fuel priming pump and Water-In-Fuel (WIF) sensor
(2) If equipped, secondary fuel filter
(3) Primary fuel filter, combined with manual fuel priming pump and Water-In-Fuel (WIF) sensor
(4) Coolant diverter valve
(5) Diesel Exhaust Fluid (DEF) pump with DEF pump filter
(6) DEF tank with DEF header installed
(7) DEF heated lines
(8) Link harness
(9) Electric control module
(10) Inlet temperature sensor
(11) Exhaust assembly
(12) NOx sensors
(13) Low-pressure fuel line

i08067930

Model View Illustrations

SMCS Code: 1000

The following model views show typical features of the engine. Due to individual applications, your engine may appear different from the illustrations.

C3.6 Industrial Engine Views For Engines Equipped With A Balancer

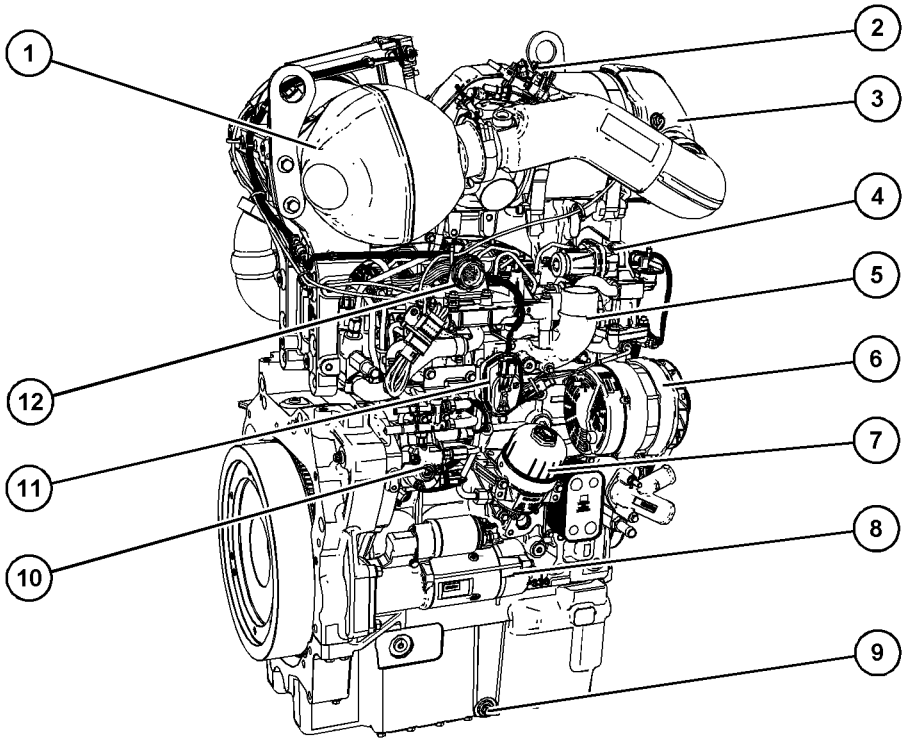


Illustration 32 g06511376

Typical example

(1) Selective Catalytic Reduction (SCR)
(2) Diesel Exhaust Fluid (DEF) injector
(3) Diesel Particulate Filter (DPF)
(4) NOx Reduction System (NRS) valve
(5) Air intake from air charge cooler
(6) Alternator
(7) Oil filter assembly
(8) Starting motor
(9) Oil drain plug (1 of 2)
(10) High-pressure fuel pump
(11) Throttle control valve
(12) Engine harness interface

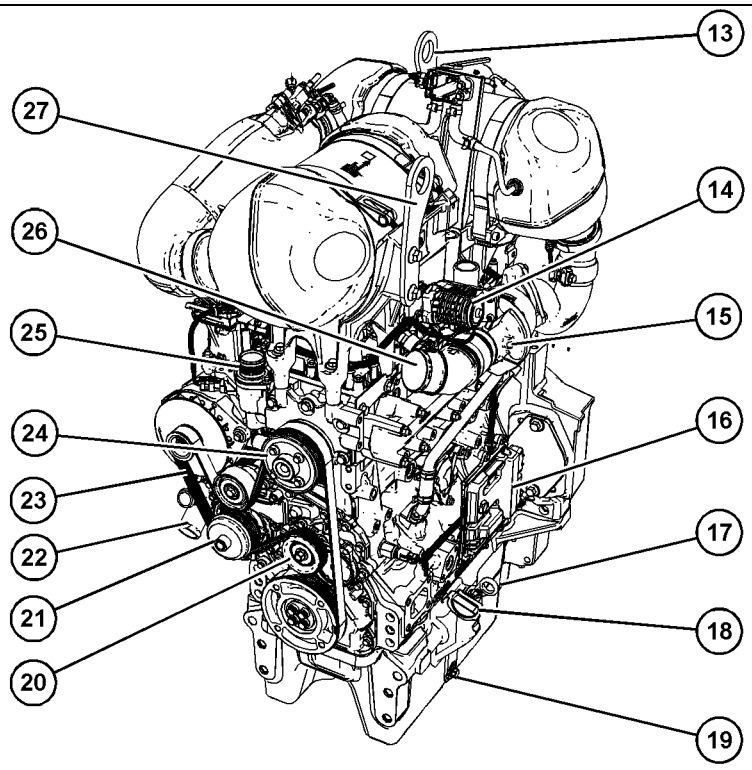


Illustration 33 g06511390

Typical example

(13) Rear lifting eyes
(14) Actuator for turbocharger
(15) Turbocharger
(16) Electronic Control Module (ECM), location for transportation only
(17) Oil gauge (Dipstick)
(18) Oil filler
(19) Oil drain plug (2 of 2)
(20) Automatic adjuster for drive belt
(21) Coolant pump pulley
(22) Coolant intake
(23) Drive belt
(24) Fan Pulley
(25) Coolant outlet
(26) Air intake from air cleaner
(27) Front lifting eye

Loose or Off Engine Components

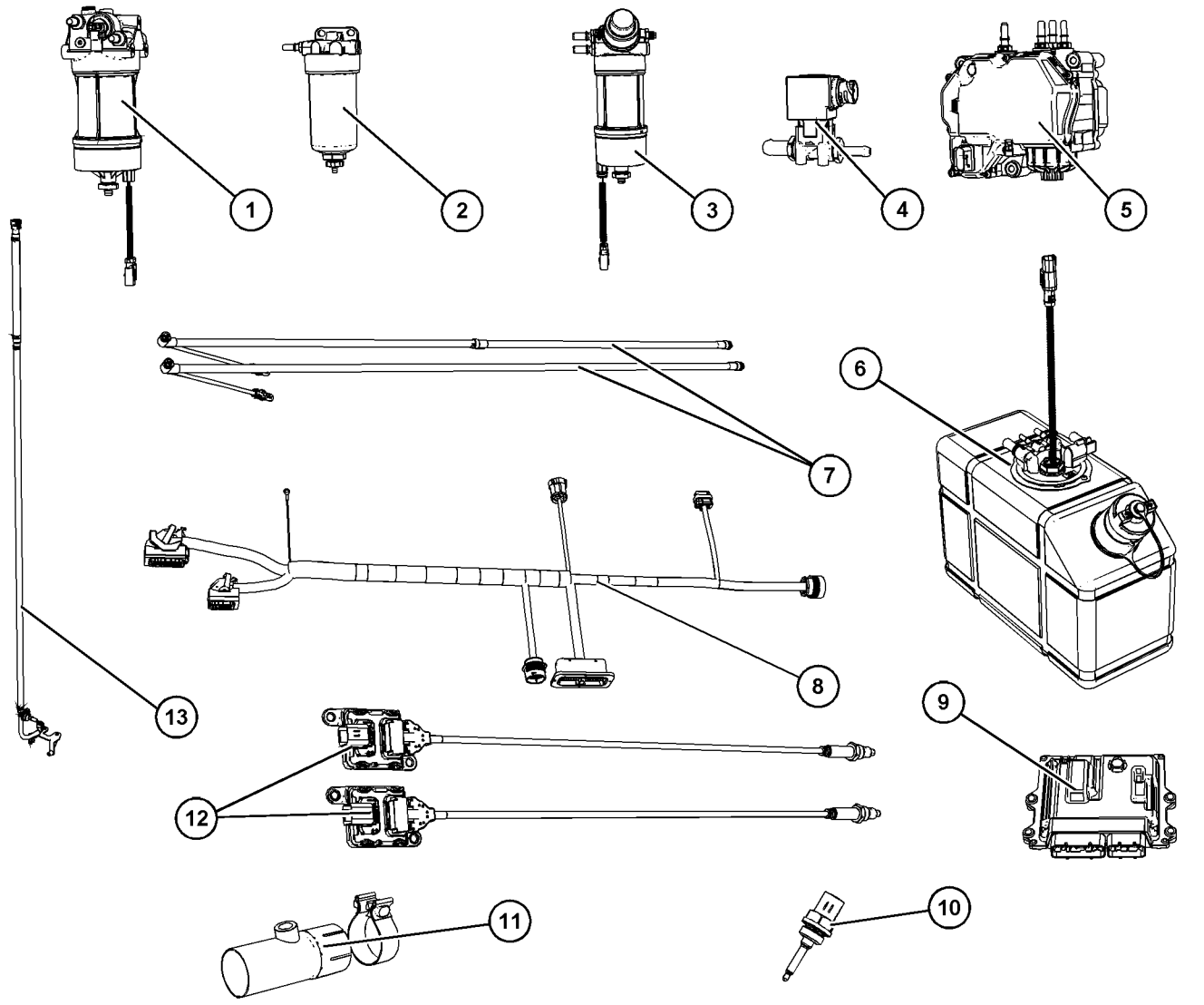


Illustration 34 g06297651

Typical example

(1) Primary fuel filter, combined with electric fuel priming pump and Water-In-Fuel (WIF) sensor
(2) If equipped, secondary fuel filter
(3) Primary fuel filter, combined with manual fuel priming pump and Water-In-Fuel (WIF) sensor
(4) Coolant diverter valve
(5) Diesel Exhaust Fluid (DEF) pump with DEF pump filter
(6) DEF tank with DEF header installed
(7) DEF heated lines
(8) Link harness
(9) Electric control module
(10) Inlet temperature sensor
(11) Exhaust assembly
(12) NOx sensors
(13) Low-pressure fuel line

i09430651

Model View Illustrations

SMCS Code: 1000

The following model views show typical features of the engine. Due to individual applications, your engine may appear different from the illustrations.

C3.6 Open Power Unit (OPU)

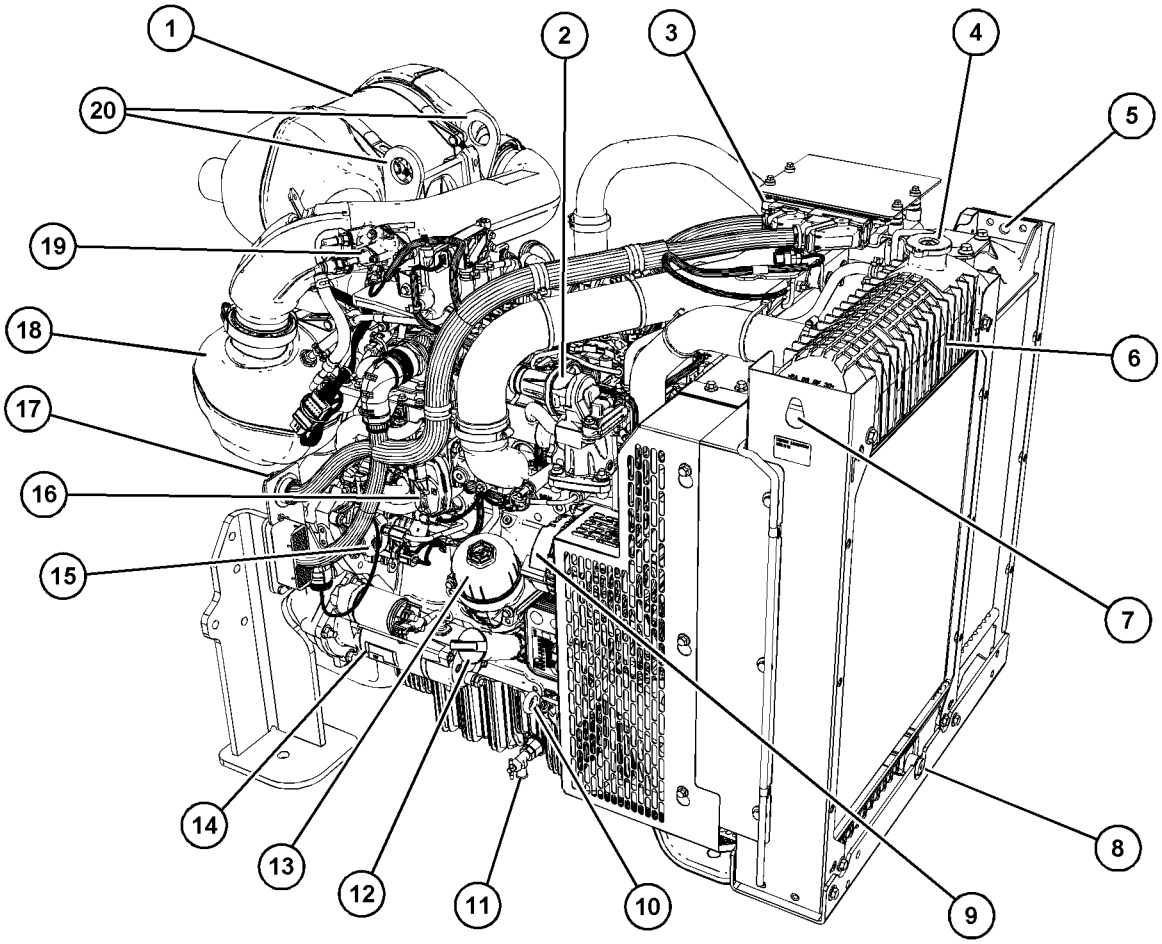


Illustration 35 g06737708

Typical example

(1) Selective Catalytic Reduction (SCR)
(2) NOx Reduction System (NRS) valve
(3) Electronic Control Module (ECM)
(4) Radiator filler cap
(5) Radiator lifting point (1 of 2)
(6) Radiator
(7) Radiator lifting point (2 of 2)
(8) Radiator drain plug
(9) Alternator
(10) Oil gauge (Dipstick)
(11) Oil drain tap
(12) Lower Oil filler
(13) Oil filter assembly
(14) Starting motor
(15) High-pressure fuel pump
(16) Throttle control valve
(17) Engine harness interface
(18) Diesel Particulate Filter (DPF)
(19) Diesel Exhaust Fluid (DEF) injector
(20) Engine Rear lifting eyes

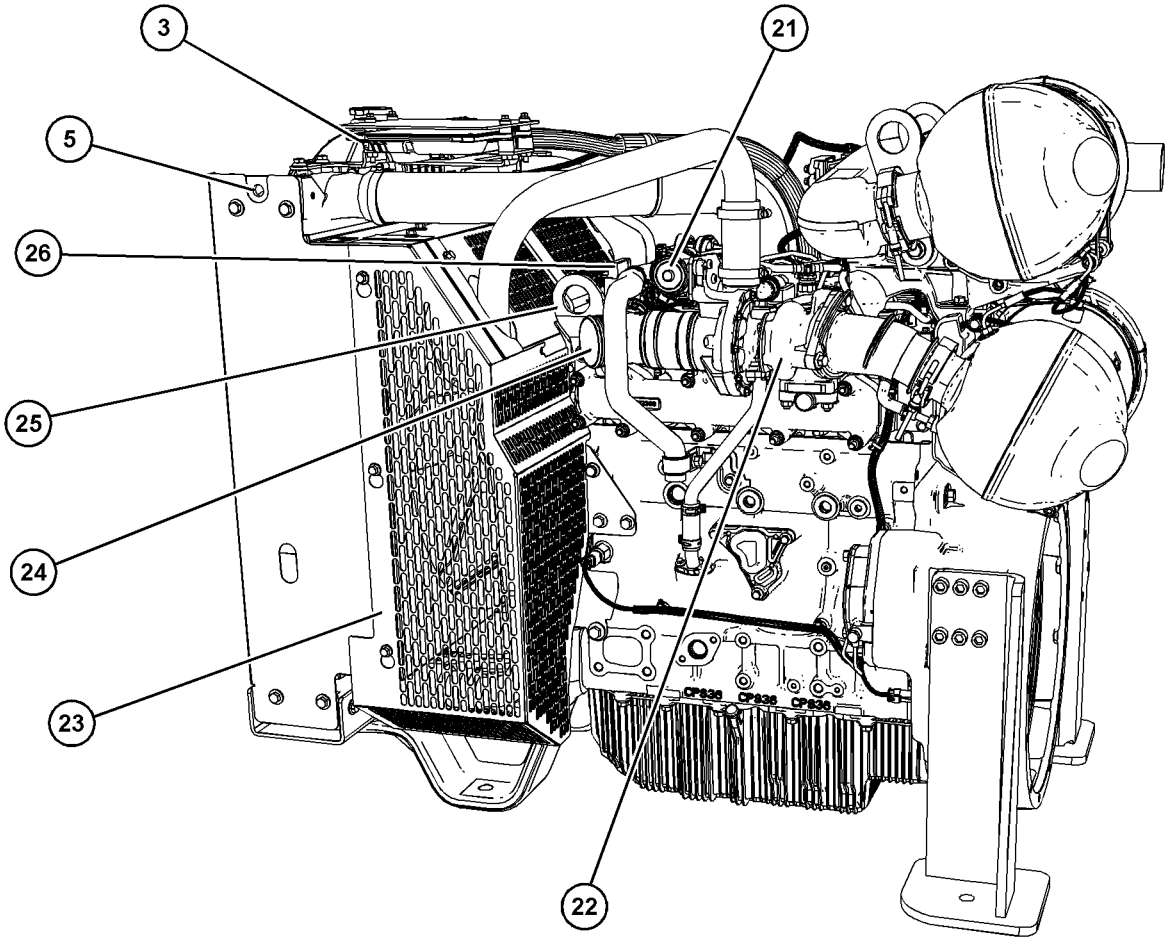


Illustration 36 g06737717

Typical example

(3) Electronic Control Module (ECM)
(5) Radiator lifting point (1 of 2)
(21) Actuator for turbocharger
(22) Turbocharger
(23) Fan guard
(24) Air intake from air cleaner
(25) Engine Front lifting eye
(26) Upper Oil filler

Loose or Off Engine Components for C3.6 OPU

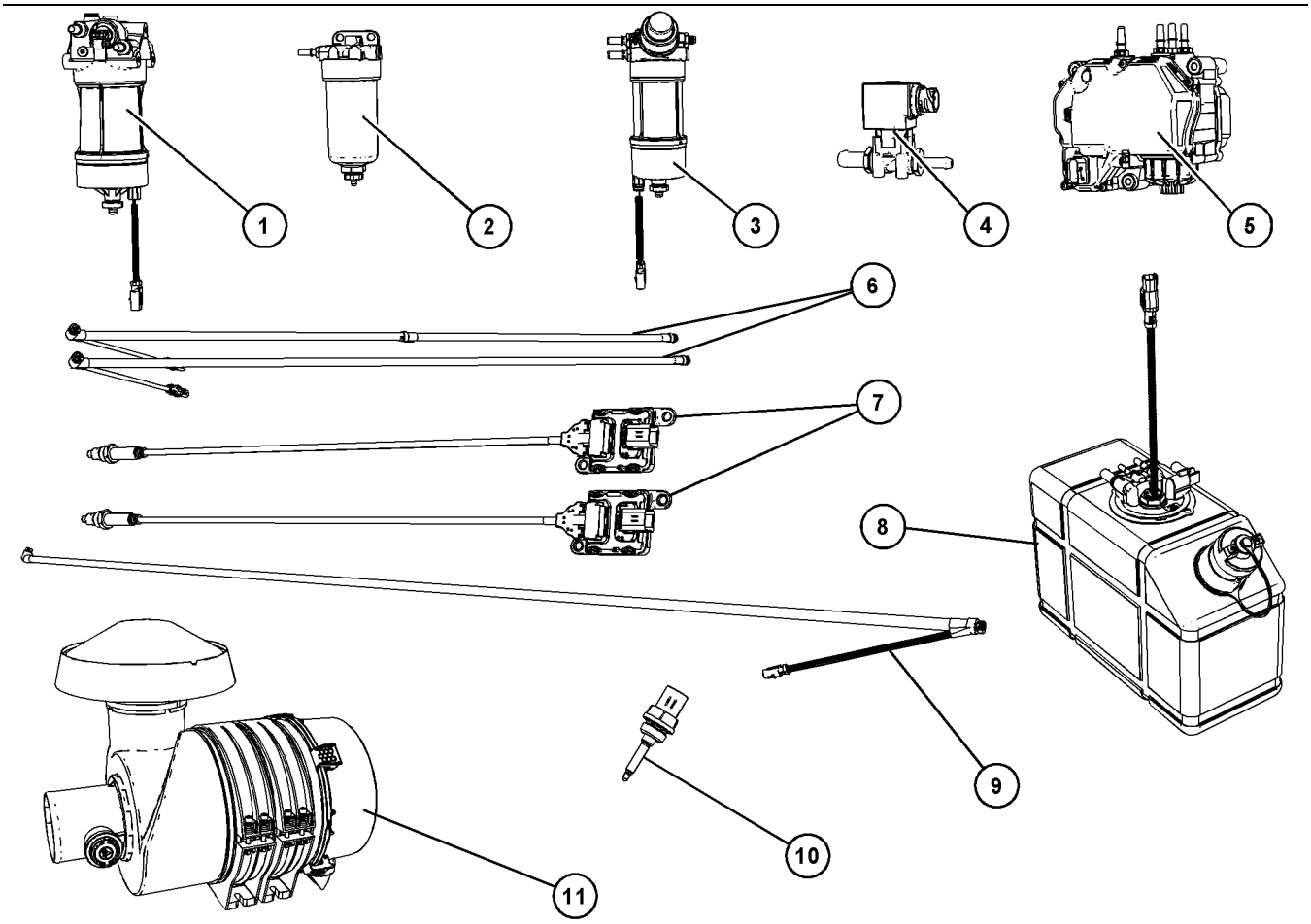


Illustration 37 g06737944

Typical example

(1) Primary fuel filter, combined with electric fuel priming pump and Water-In-Fuel (WIF) sensor
(2) If equipped, secondary fuel filter
(3) Primary fuel filter, combined with manual fuel priming pump and Water-In-Fuel (WIF) sensor
(4) Coolant diverter valve
(5) Diesel Exhaust Fluid (DEF) pump with DEF pump filter
(6) DEF heated lines
(7) NOx sensors
(8) DEF tank with DEF header installed
(9) line
(10) Inlet temperature sensor
(11) Engine air cleaner

C2.8 OPU

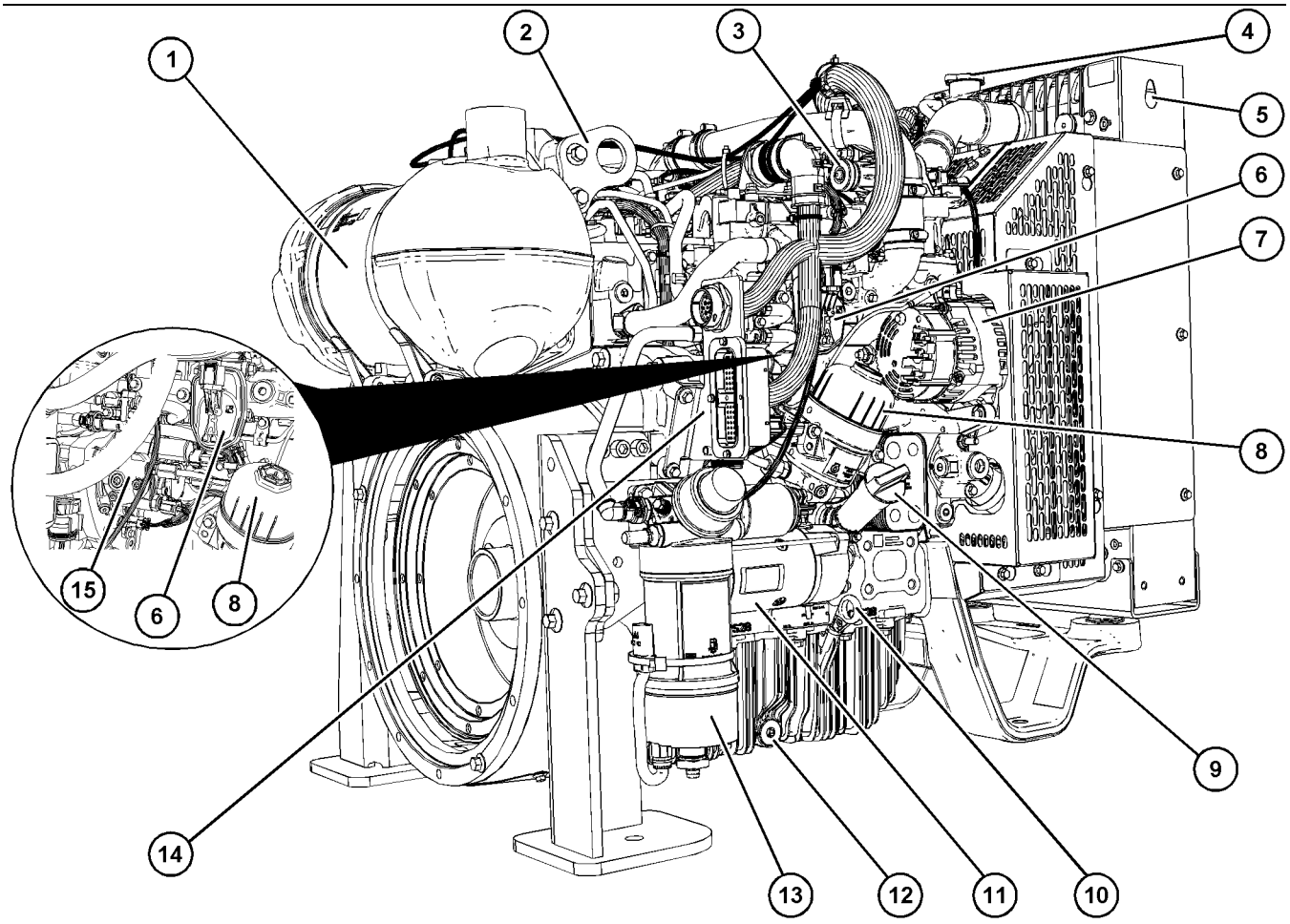


Illustration 38 g06740506

Typical example

(1) Diesel Particulate Filter (DPF) and Diesel Oxidation Catalyst (DOC) combined
(2) Engine Rear lifting eye
(3) NOx Reduction System (NRS) valve
(4) Radiator filler cap
(5) Radiator lifting point (1 of 2)
(6) Throttle control valve
(7) Alternator
(8) Oil filter assembly
(9) Lower Oil filler
(10) Oil gauge (Dipstick)
(11) Starting motor
(12) Oil drain plug
(13) Primary fuel filter
(14) Engine harness interface
(15) High-pressure fuel pump

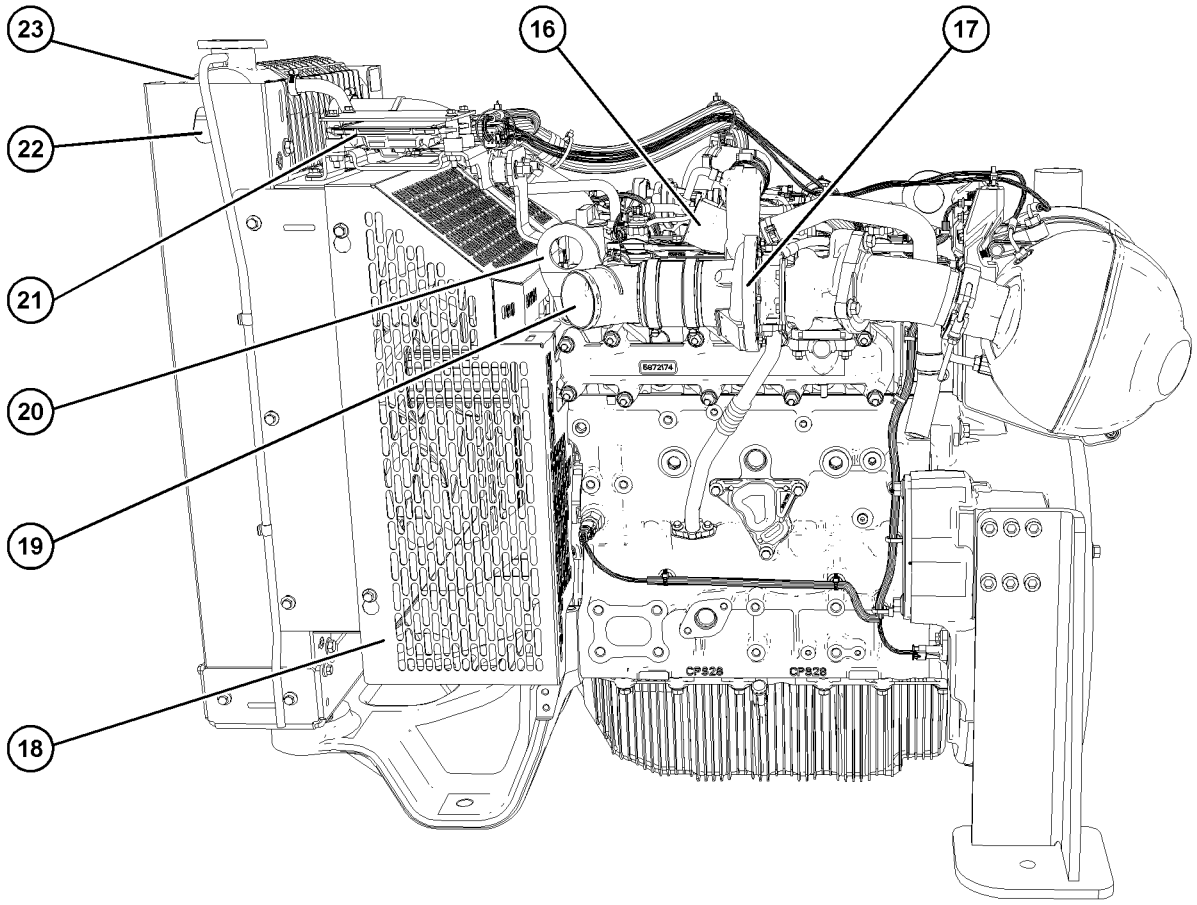


Illustration 39 g06740500

Typical example

(16) Actuator for turbocharger
(17) Turbocharger
(18) Fan guard
(19) Air intake from air cleaner
(20) Engine Front lifting eye
(21) Upper Oil filler
(22) Electronic Control Module (ECM)
(23) Radiator lifting point (2 of 2)

Loose or Off Engine Components for C2.8 OPU

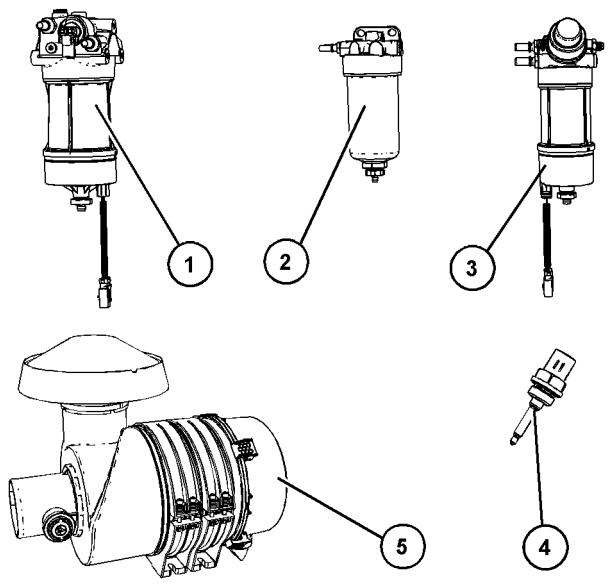


Illustration 40 g06740514

Typical example

(1) Primary fuel filter, combined with electric fuel priming pump and Water-In-Fuel (WIF) sensor
(2) If equipped, secondary fuel filter
(3) Primary fuel filter, combined with manual fuel priming pump and Water-In-Fuel (WIF) sensor
(4) Inlet temperature sensor
(5) Engine air cleaner

i07853928

Product Description
((Engine and Aftertreatment))

SMCS Code: 1000; 4450; 4491

C3.6 Industrial Engine

The Caterpillar C3.6 industrial engine is a single turbocharged, air to air charge cooled engine. There are two variants of the C3.6 industrial engines. Engines with a power rating of 56 kW (75.1 hp) and below or engines with a power rating of 70 kW (93.9 hp) and above.

The engine aftertreatment system for engines with a power ratings of 56 kW (75.1 hp) and below, is configured as follows:

- Diesel Oxidation Catalyst (DOC)
- Diesel Particulate Filter (DPF)

The engine aftertreatment system for engines with a power ratings of 70 kW (93.9 hp) and above, is configured as follows:

- Diesel Oxidation Catalyst (DOC)
- Diesel Particulate Filter (DPF)
- Selective Catalytic Reduction (SCR)

C2.8 Industrial Engine

The Caterpillar C2.8 industrial engine is a single turbocharged engine. The engine may be turbocharged or turbocharged with an air charge cooler.

The engine aftertreatment system is configured as follows:

- Diesel Oxidation Catalyst (DOC)
- Diesel Particulate Filter (DPF)

Engine Specifications

Note: The front end of the engine is opposite the flywheel end of the engine. The left and the right sides of the engine are determined from the flywheel end. The number 1 cylinder is the front cylinder.

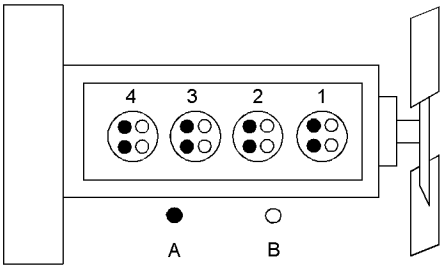


Illustration 41 g06297997

Typical example

(A) Inlet valves
(B) Exhaust valves

Table 1

C3.6 Engine Specifications	
Operating Range (rpm)	**800 to 2400** (1)
Number of Cylinders	4 In-Line
Bore	98 mm (3.86 inch)
Stroke	120 mm (4.72 inch)
Power rating (engines with DOC and DPF)	55.4 kW (74.29 hp)

(continued)

(Table 1, contd)

Power rating (engines with DOC, DPF, and SCR)	70 to 100 kW (93.87 to 134.1 hp)
Aspiration	Turbocharged charge cooled
Compression Ratio	17:1
Displacement	3.621 L (220.966 cubic inch)
Firing Order	1-3 -4-2
Valves per cylinder	4
Rotation (flywheel end)	Counterclockwise

(1) The operating rpm depends on the engine rating, the application, and the configuration of the throttle.

Table 2

C2.8 Engine Specifications	
Operating Range (rpm)	**800 to 2400** (1)
Number of Cylinders	4 In-Line
Bore	90 mm (3.54 inch)
Stroke	110 mm (4.33 inch)
Power rating	50 kW to 55.4 kW (67.05 hp to 74.29 hp)
Aspiration	Turbocharged and Turbocharged charge cooled
Compression Ratio	17:1
Displacement	2.8 L (170.87 cubic inch)
Firing Order	1-3 -4-2
Valves per cylinder	4
Rotation (flywheel end)	Counterclockwise

(1) The operating rpm depends on the engine rating, the application, and the configuration of the throttle.

Electronic Engine Features

The engine and aftertreatment operating conditions are monitored. The Electronic Control Module (ECM) controls the response of the engine to these conditions and to the demands of the operator. These conditions and operator demands determine the precise control of fuel injection by the ECM. The electronic engine control system provides the following features:

- Engine monitoring
- Engine speed governing
- Control of the injection pressure
- Cold start strategy
- Automatic air/fuel ratio control
- Torque rise shaping
- Injection timing control
- System diagnostics
- NOx reduction system control
- Aftertreatment system control

The ECM provides an electronic governor that controls the injector output to maintain the desired engine speed.

For more information on electronic engine features, refer to the Operation and Maintenance Manual, “Features and Controls” topic (Operation Section).

Engine Diagnostics

The engine has built-in diagnostics to ensure that the engine systems are functioning correctly. The operator will be alerted to the condition by a “Stop or Warning” lamp. Under certain conditions, the engine horsepower and the vehicle speed may be limited. The electronic service tool may be used to display the diagnostic codes.

There are three types of diagnostic codes: active, logged, and event.

Most of the diagnostic codes are logged and stored in the ECM. For additional information, refer to the Operation and Maintenance Manual, “Engine Diagnostics” topic (Operation Section).

Engine Cooling and Lubrication

The cooling system and lubrication system consists of the following components:

- Belt driven water pump
- Water temperature regulator which regulates the engine coolant temperature
- Gear-driven gerotor type oil pump
- Oil cooler

The engine lubricating oil is supplied by a gerotor type oil pump. The engine lubricating oil is cooled and the engine lubricating oil is filtered. The bypass valve can provide unrestricted flow of lubrication oil to the engine if the oil filter element should become plugged.

Engine efficiency, efficiency of emission controls, and engine performance depend on adherence to proper operation and maintenance recommendations. Engine performance and efficiency also depend on the use of recommended fuels, lubrication oils, and coolants. Refer to this Operation and Maintenance Manual, “Maintenance Interval Schedule” for more information on maintenance items.

Engine Service Life

Engine efficiency and maximum utilization of engine performance depend on the adherence to proper operation and maintenance recommendations. In addition, use recommended fuels, coolants, and lubricants. Use the Operation and Maintenance Manual as a guide for required engine maintenance.

Expected engine life is generally predicted by the average power that is demanded. The average power that is demanded is based on fuel consumption of the engine over time. Reduced hours of operation at full throttle and/or operating at reduced throttle settings result in a lower average power demand. Reduced hours of operation will increase the length of operating time before an engine overhaul is required. For more information, refer to the Operation and Maintenance Manual, “Overhaul Considerations” topic (Maintenance Section).

Aftermarket Products and Caterpillar Engines

Caterpillar does not warrant the quality or performance of non-Caterpillar fluids and filters.

When auxiliary devices, accessories, or consumables (filters, additives, catalysts), which are made by other manufacturers are used on Caterpillar products, the Caterpillar warranty is not affected simply because of such use.

However, failures that result from the installation or use of other manufacturers devices, accessories, or consumables are NOT Caterpillar defects. Therefore, the defects are NOT covered under the Caterpillar warranty.

Aftertreatment System

The aftertreatment system is approved for use by Caterpillar. To be emissions-compliant, only the approved Caterpillar aftertreatment system must be used on a Caterpillar engine.

Product Identification Information

i07816292

Plate Locations and Film Locations

SMCS Code: 1000; 4450

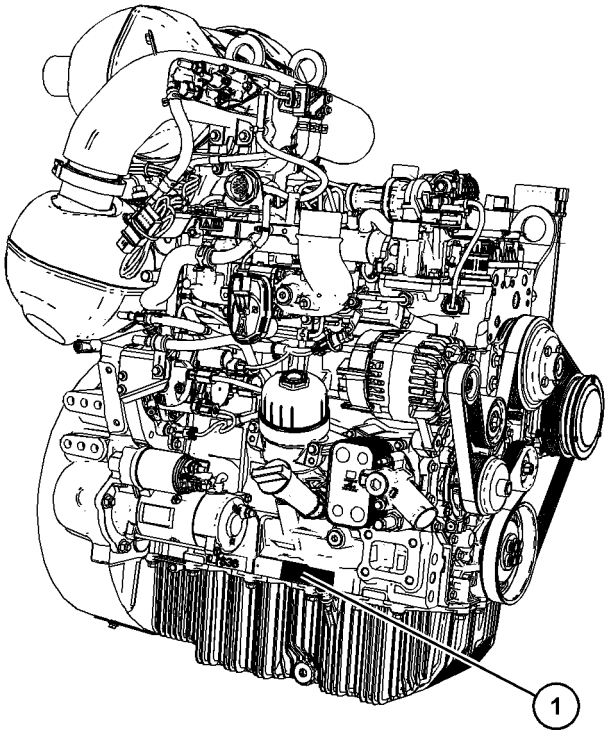


Illustration 42 g06298126

Typical example

(1) Engine serial number plate location

Serial Number Plate

The engine serial number plate (1) is on the right side of the cylinder block to the rear of the front engine mounting.

Caterpillar dealers need all these numbers to determine the components that were included with the engine. This information permits accurate identification of replacement part numbers.

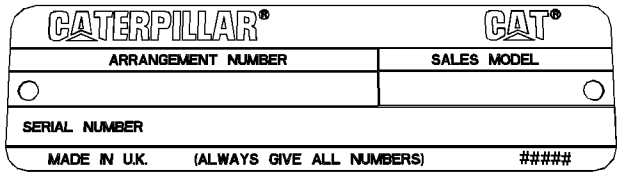


Illustration 43 g03047463

Typical example of serial number plate

i08043623

Emissions Certification Film

SMCS Code: 1000; 7405

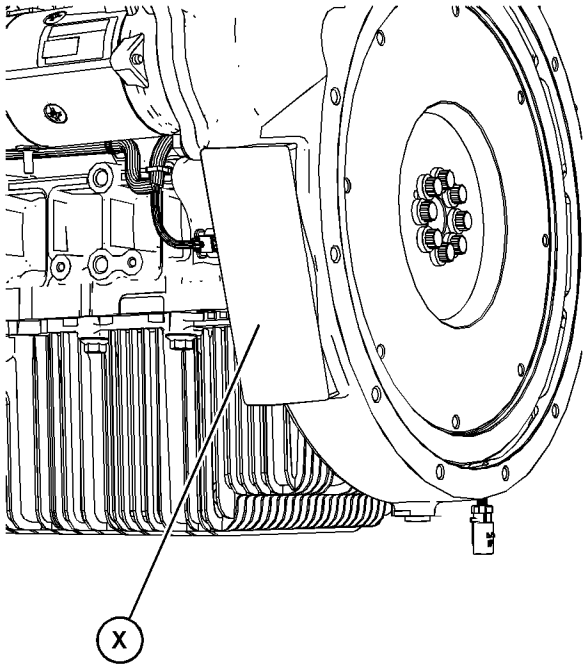


Illustration 44 g06522161

Typical location of the engine emission label

The engine emission label is typically installed on the flywheel housing at Position (X). An extra engine emission label may be supplied loose.

i07925928

Reference Information

SMCS Code: 1000; 4450

Information for the following items may be needed to order parts. Locate the information for your engine. Record the information in the appropriate space. Make a copy of this list for a record. Keep the information for future reference.

Record for Reference

Engine Model _____

Engine Serial number _____

Engine Low Idle rpm _____

Engine Full Load rpm _____

In Line Fuel Filter _____

Primary Fuel Filter _____

Secondary Fuel Filter Element _____

Lubrication Oil Filter Element _____

Auxiliary Oil Filter Element (if equipped)_____

Total Lubrication System Capacity _____

Total Cooling System Capacity _____

Air Cleaner Element _____

Drive Belt _____

Clean Emissions Module

Part Number _____

Serial Number _____

Diesel Exhaust Fluid (DEF) Pump

Part Number _____

Serial Number _____

Operation Section

Lifting and Storage

i09512670

Product Lifting

SMCS Code: 7000; 7002

NOTICE
Always inspect lifting eyebolts and all other lifting equipment for damage before performing any lifting. Never bend the eyebolts and the brackets. Never perform product lifting if components are damaged. Only load the eyebolts and the brackets under tension. Remember that the capacity of an eyebolt is less as the angle between the supporting members and the object becomes less than 90 degrees.

When it is necessary to remove a component at an angle, only use a link bracket that is properly rated for the weight.

Read all the information within product lifting before any lifting is attempted. Ensure that the correct set of lifting eyes for the assembly to be lifted have been selected.

Use a hoist to remove heavy components. Use an adjustable lifting beam to lift the engine. All supporting members (chains and cables) should be parallel to each other. The chains and cables should be perpendicular to the top of the object that is being lifted.

To obtain the correct balance for lifting an application, adjust the chain lengths.

Lifting eyes are designed and installed for the specific engine arrangement. Alterations to the lifting eyes and/or the engine, make the lifting eyes and the lifting fixtures obsolete. If alterations are made, ensure that proper lifting devices are provided.

There are several different designs of lifting eyes. The following sections give examples of lifting eyes on the engine and engine with aftertreatment.

Consult your Caterpillar dealer for information regarding fixtures for proper engine lifting.

C3.6 Industrial Engine

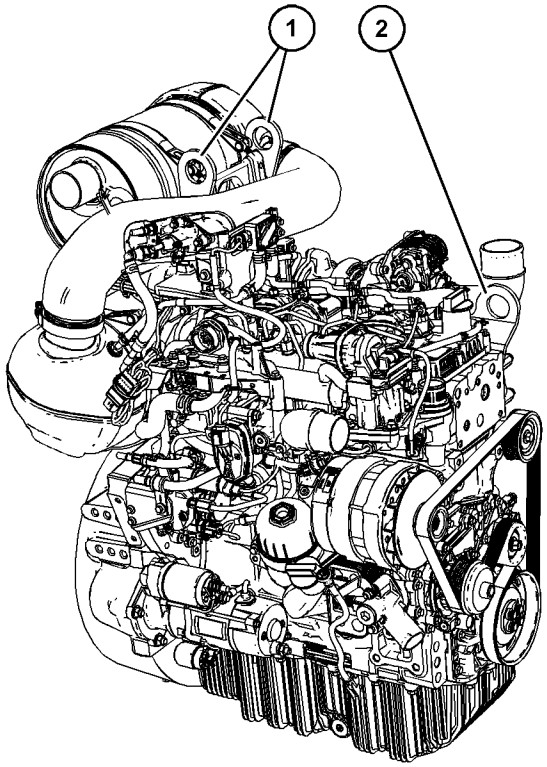


Illustration 45 g06298473

Typical example

(1) Rear lifting eyes
(2) Front lifting eye

Use all 3 lifting eyes when lifting engine assembly.

Caterpillar recommends that model-specific cradles are used to store and transport engines. Supporting engines on the sump/oil pan is not recommended under any circumstances. This action is for both safety and quality reasons.

C3.6 Industrial Engine Equipped With A Balancer

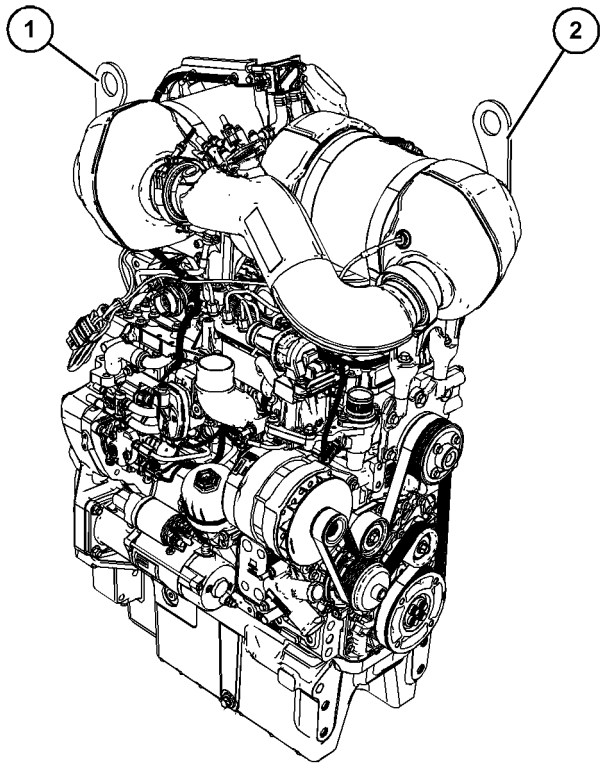


Illustration 46 g06511783

Typical example

(1) Rear lifting eyes
(2) Front lifting eye

Use both lifting eyes when lifting engine assembly.

Caterpillar recommends that model-specific cradles are used to store and transport engines. Supporting engines on the sump/oil pan is not recommended under any circumstances. This action is for both safety and quality reasons.

C3.6 Open Power Unit (OPU)

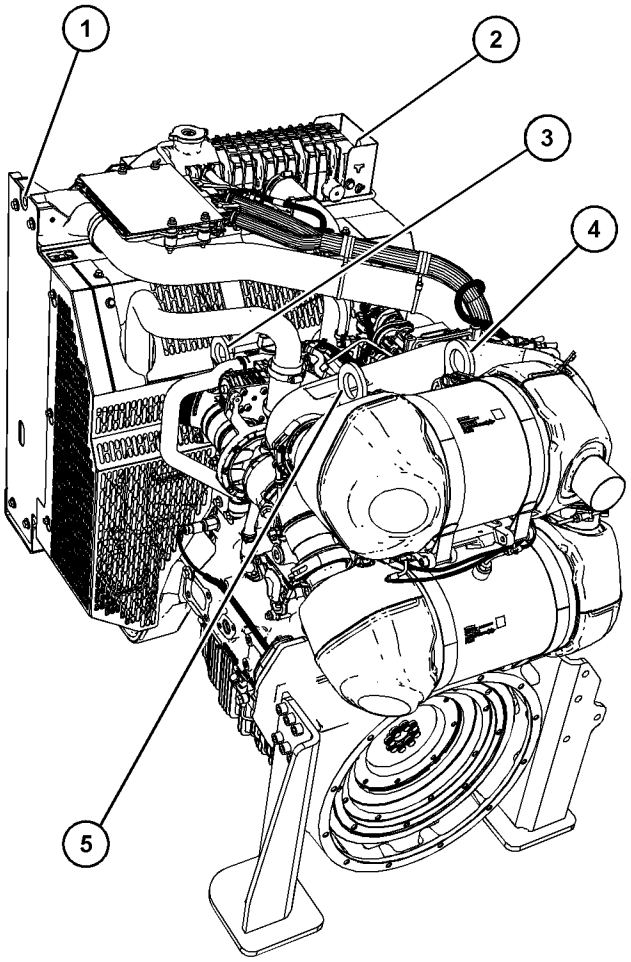


Illustration 47 g06742363

Typical example

(1) Radiator lifting eye (1 of 2)
(2) Radiator lifting eye (2 of 2)
(3) Front lifting eye
(4) Rear lifting eye
(5) Rear lifting eye

Use all 3 lifting eyes when lifting engine assembly.

Use both lifting eyes when lifting radiator assembly.